

Lesson 3 of 5

Modeling and Analysis

Intro to R workshop, LU Skill School

Instructor:

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Slides adapted from and inspired by:
Irina Vartanova, Institute for Future Studies, Stockholm

Today's agenda

- Aggregating Data
- Grouping Cases
- Descriptives
- Bivariate Analyses
- Multivariate Analyses
- Quadratic Terms
- Interaction Terms

- Use the link to download the files we will be using today:
https://github.com/ChristopherSwader/R_introduction
- Download the Day3 folder.

Aggregate

- Clear your environment
- Set your working directory to Day 3.
- Load the moral_issues data from day 2!
 - Either navigate through Rstudio's import menu and paste the code
 - Or if you know the code but just need the path, you can get it through the files pane → settings

summarise()

Compute table of summaries.

Alternative for one summary stat:

```
aggregate(moral_issues$n,  
by=list(issue=moral_issues$issue), sum)
```

```
moral_issues %>%  
  group_by(issue) %>%  
  summarise(total = sum(n), max = max(n))
```

moral_issues

issue	year	n	prop
fehome	1990	890	0.8213483
libath	1990	881	0.6912599
marblk	1990	940	0.1670213
polescap	1990	854	0.2236534
spkrac	1990	892	0.6423767
fehire	1996	1236	0.6480583



	issue	total	max
	<chr>	<dbl>	<dbl>
1	abany	36794	1939
2	abdefect	44152	1926
3	abhlth	44360	1934
4	abnomore	44072	1925
5	abpoor	44023	1928
6	abrape	43922	1934
7	absingle	44020	1938
8	advfront	8559	1806
9	affrmact	18944	1904
10	busing	22886	1859

Your Turn

*Alter the last code to extract the rows where **issue == "abany"**. Then use **summarise()** and **mean()**, **min()**, and **max()** to find:*

- 1. The average public opinion for the issue over all time points it was measured.*
- 2. The first and the last **years** the issue appeared in the survey.*

```
moral_issues %>%  
  filter(issue == "abany") %>%  
  summarise(mean_prop = mean(prop),  
            first = min(year),  
            last = max(year))
```

```
# A tibble: 1 × 3
```

	mean_prop	first	last
	<dbl>	<dbl>	<dbl>
1	0.414	1977	2018

n()

The number of rows in a dataset/group

```
moral_issues %>% summarise(n = n())
```

moral_issues

issue	year	n	prop
fehhome	1990	890	0.8213483
libath	1990	881	0.6912599
marblk	1990	940	0.1670213
polescap	1990	854	0.2236534
spkrac	1990	892	0.6423767
fehired	1996	1236	0.6480583



n
1651

A more simple alternative:

nrow(moral_issues)

n_distinct()

The number of distinct values in a variable

```
moral_issues %>% summarise(n = n(),  
                           n_issue = n_distinct(issue))
```

moral_issues

issue	year	n	prop
fehhome	1990	890	0.8213483
libath	1990	881	0.6912599
marblk	1990	940	0.1670213
polescap	1990	854	0.2236534
spkrac	1990	892	0.6423767
fehfire	1996	1236	0.6480583



n	n_issue
1651	81

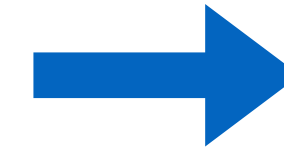
A base R alternative:

length(unique(moral_issues\$issue))

More on
grouping
cases

Here: grouping by just one variable (city)

city	particle size	amount ($\mu\text{g}/\text{m}^3$)
New York	large	23
New York	small	14



mean	sum	n
18,5	37	2

London	large	22
London	small	16



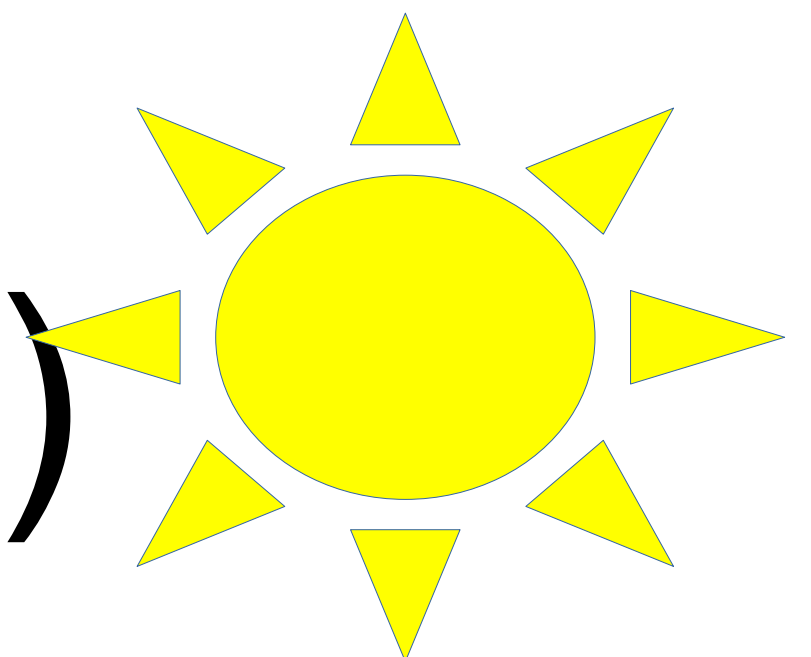
19,0	38	2
------	----	---

Beijing	large	121
Beijing	small	56



88,5	177	2
------	-----	---

`group_by() + summarise()`



group_by()

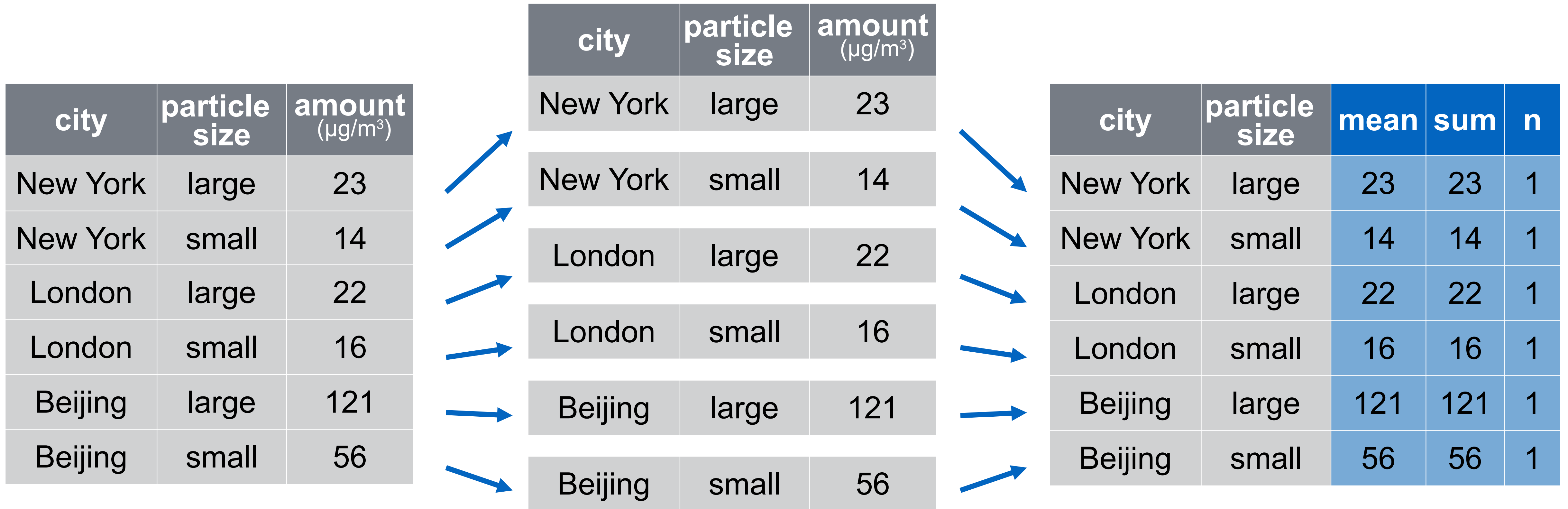
Groups cases by common values of one or more columns.

```
pollution %>%  
  group_by(city)
```

```
# A tibble: 6 x 3  
# Groups:   city [3]  
  city      size amount  
  <chr>    <chr>  <dbl>  
1 New York large    23  
2 New York small    14  
3 London   large    22
```

*They are grouped,
but no operations are
yet performed on the
cases as groups.*

group_by() with multiple grouping variables



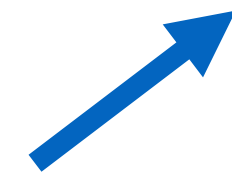
```
pollution %>%  
  group_by(city, size) %>%  
  summarise(mean = mean(amount), sum = sum(amount), n = n())
```

Your Turn

Use `group_by()` and `summarize` to get the `mean()` the `sum()` and the `n()` of each city in the toy pollution dataset.

group_by()

city	particle size	amount ($\mu\text{g}/\text{m}^3$)
New York	large	23
New York	small	14
London	large	22
London	small	16
Beijing	large	121
Beijing	small	56



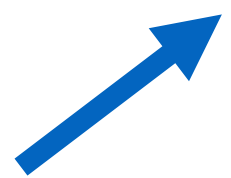
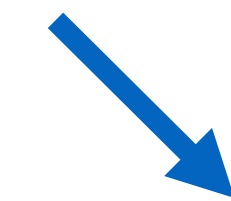
city	particle size	amount ($\mu\text{g}/\text{m}^3$)
New York	large	23
New York	small	14



London	large	22
London	small	16



Beijing	large	121
Beijing	small	56



city	mean	sum	n
New York	18,5	37	2
London	19,0	38	2
Beijing	88,5	177	2

```
pollution %>%
```

```
  group_by(city) %>%
```

```
  summarise(mean = mean(amount), sum = sum(amount), n = n())
```

ungroup()

Removes grouping criteria from a data frame.

```
pollution %>%  
  group_by(city) %>%  
  ungroup() %>%  
  summarise(sum = sum(amount))
```

```
pollution %>%  
  group_by(city) %>%  
  summarise(sum = sum(amount))
```

sum
252

city	sum
New York	37
London	38
Beijing	177

Your Turn

*With `moral_issues`, use **`group_by()`**, **`summarise()`**, and **`arrange()`** to display the ***issues*** with the highest average public opinion.*

```
moral_issues %>%  
  group_by(issue) %>%  
  summarise(mean_prop = mean(prop)) %>%  
  arrange(desc(mean_prop))
```

```
#   issue      mean_prop  
# 1 hitmarch    0.967  
# 2 racfew      0.952  
# 3 marwht      0.943  
# 4 hitdrunk    0.915  
# 5 polmurdr    0.901  
# 6 abhlth      0.896  
# 7 polabuse     0.886  
# ... with 71 more rows
```

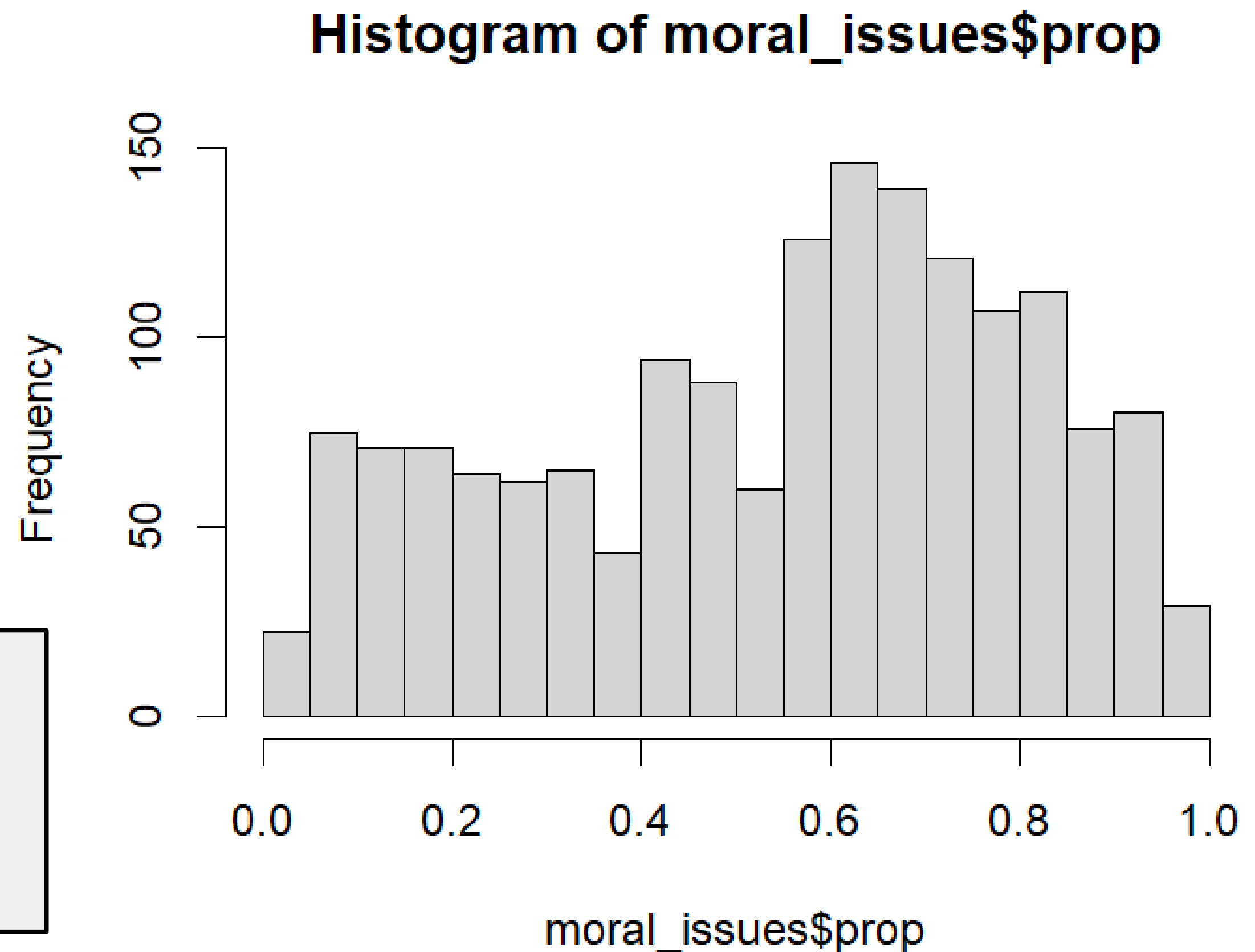
Descriptive stats

Descriptives of one (numerical) variable

```
summary(moral_issues$prop)
mean(moral_issues$prop)
min(moral_issues$prop)
max(moral_issues$prop)
sd(moral_issues$prop)

hist(moral_issues$prop, breaks=20)
```

This histogram version is the basic one. GGplot2 can produce much more fancy versions. Day 4



Descriptives of one (categorical) variable

```
count(moral_issues, issue) #tidyverse way  
table(moral_issues$issue) #base R way
```

Descriptives of all variables:

Can i just get a quick snapshot?

Your Turn

Install the package "skimr". Load it. Then run the function skim() on moral_issues.

skim() output: Look at whole dataset in a glance

```
> skim(moral_issues)
```

— Data Summary

	Values
Name	moral_issues
Number of rows	1651
Number of columns	6

Column type frequency:

character	1
numeric	5

Group variables

None

— Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
1 issue	0	1	5	8	0	81	0

— Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
1 year	0	1	1994.	12.9	1972	1985	1993	2006	2018	
2 n	0	1	1348.	386.	395	986	1418	1653	2815	
3 prop	0	1	0.544	0.256	0.0150	0.338	0.595	0.748	0.978	
4 prop_lib	54	0.967	0.627	0.255	0.00990	0.450	0.705	0.828	0.984	
5 prop_cons	54	0.967	0.491	0.262	0.0173	0.258	0.539	0.692	0.989	

>

Bivariate relationships

Correlation Tables

```
cor(na.omit(moral_issues)[,-1]) #
```

Can you explain my code?

*You can specify
method="spearman" for
ordinal data*

Correlation Tables with signif levels

```
install.packages("corrtable")

library(corrtable)

correlation_matrix(moral_issues, digits = 2 , use = "lower",
replace_diagonal = T)
```

	year	n	prop	prop_lib	prop_cons
year	""	""	""	""	""
n	" 0.02	" ""	""	""	""
prop	" 0.05*	" "-0.05*	" ""	""	""
prop_lib	" 0.10***"	" "-0.05*	" " 0.98***"	""	""
prop_cons	"-0.01	" "-0.03	" " 0.98***"	" 0.93***"	""

Correlation Heatmaps in R

```
library(reshape2)
upper_tri <- my_corr_matrix * upper.tri(my_corr_matrix, diag = T)

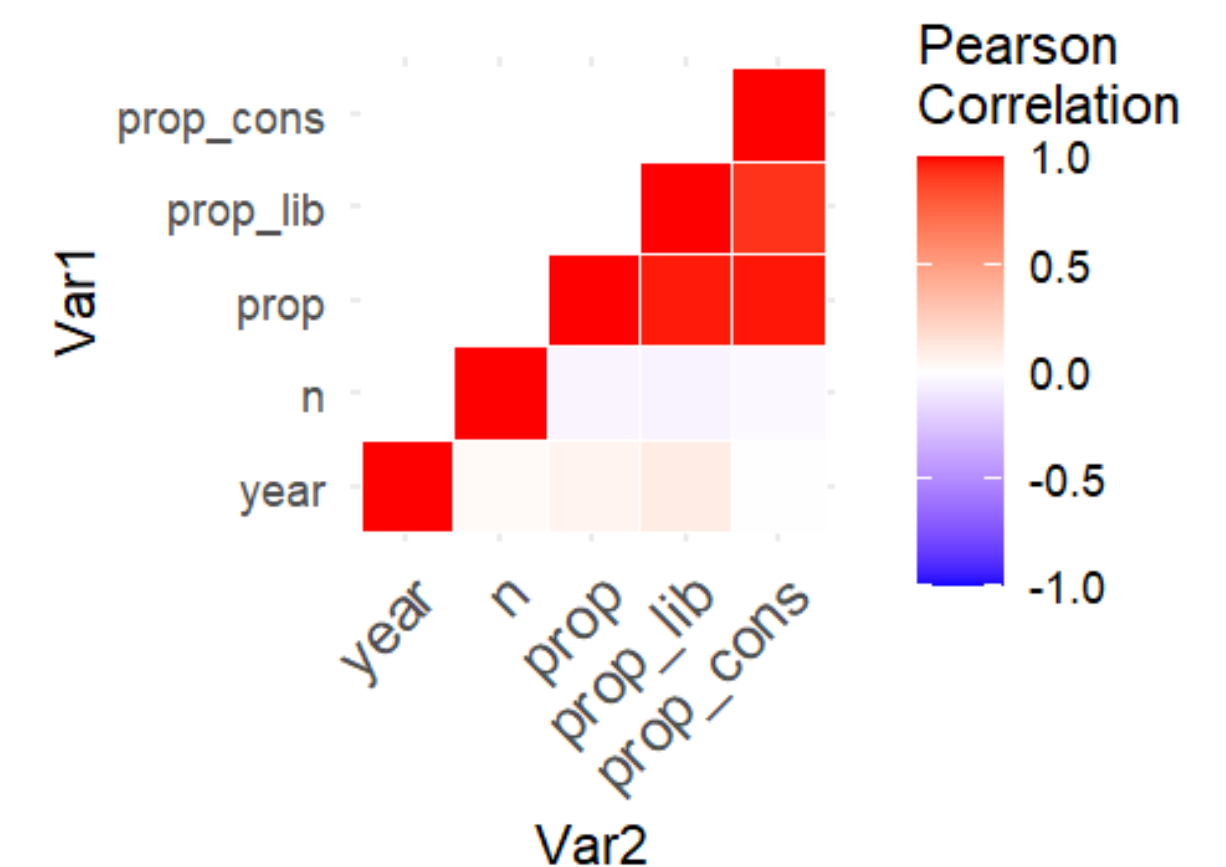
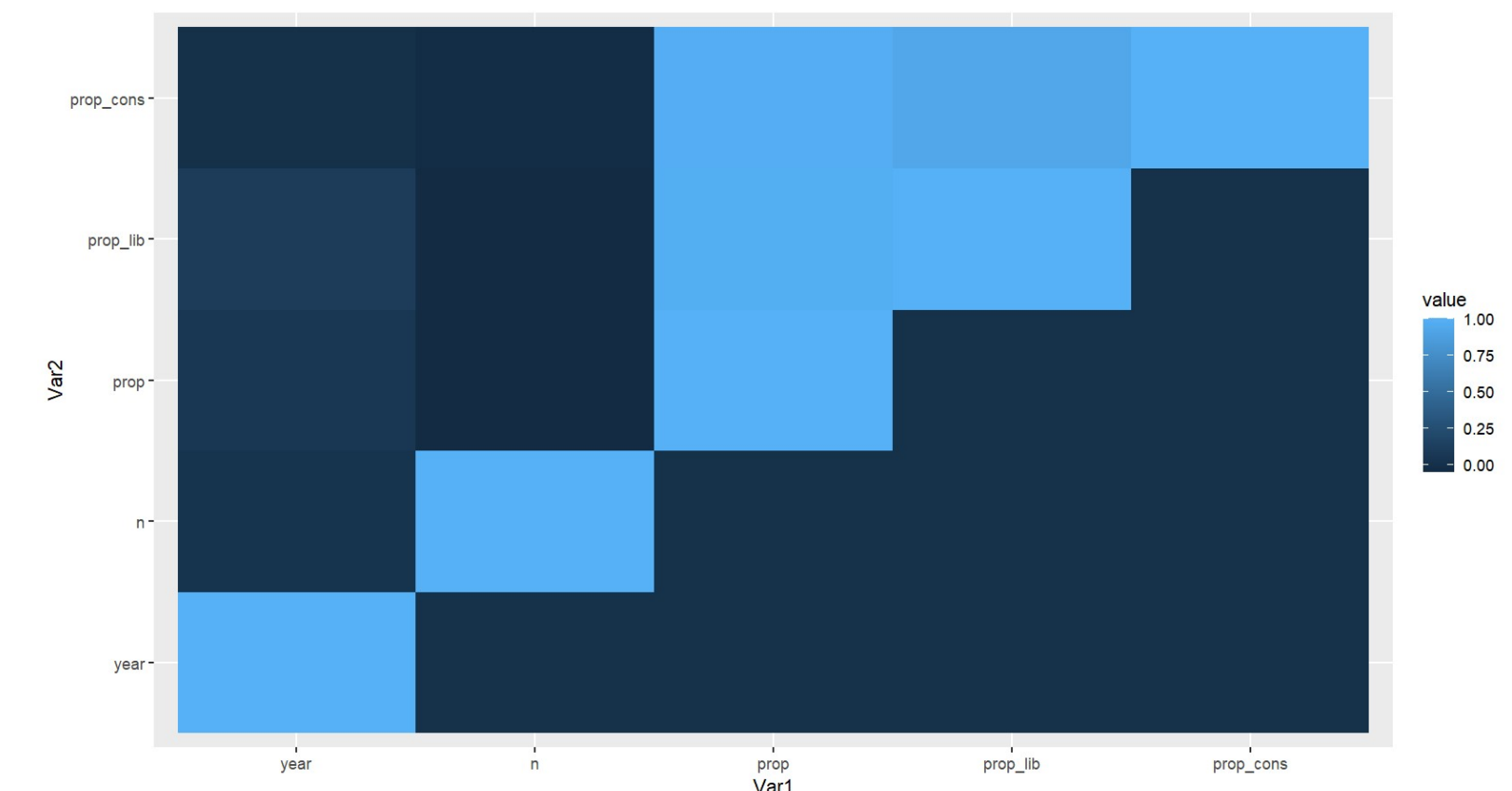
my_corr_matrix <- cor(na.omit(moral_issues)[, -1])

melted_matrix <- melt(upper_tri)
library(ggplot2)

ggplot(data = melted_matrix, aes(x=Var1, y=Var2, fill=value)) +
  geom_tile()

ggplot(data = melted_matrix, aes(Var2, Var1, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
    midpoint = 0, limit = c(-1,1), space = "Lab",
    name="Pearson\nCorrelation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1,
    size = 12, hjust = 1)) +
  coord_fixed()
```

Try
<https://r-graph-gallery.com/>
for a huge range of R resources



Bi- and Multi-variate Modeling

Some traditional modelling functions in R

function	package	fits
lm()	stats	linear models
glm()	stats	generalized linear models
gam()	mgcv	generalized additive models
rlm()	MASS	robust linear models
lmer()	lme4	linear mixed-effects models

Also consider machine learning methods, all fully accessible via R: random forest, XGboost, artificial neural nets, topic modeling, etc.

(Popular) modelling functions in R

function	package	fits
lm()	stats	linear models
glm()	stats	generalized linear models
gam()	mgcv	generalized additive models
rlm()	MASS	robust linear models
lmer()	lme4	linear mixed-effects models

```
flfp #female labor force participation
```

cntry <chr>	region <fctr>	wvs_flfp <dbl>	patr_mean <dbl>	log_gdp <dbl>
Algeria	mena	40.848806	1.31699822	9.528154
Armenia	centr asia	54.267245	0.55224440	8.825190
Australia	west	86.936284	-1.13341284	10.531036
Azerbaijan	centr asia	61.691654	1.33496402	9.664859
Belarus	s/est europe	86.852032	0.09008820	9.717362
Brazil	latin	67.250674	-0.79309932	9.513073
Bulgaria	s/est europe	90.073361	-0.43430856	9.222664
Burkina Faso	ss africa	41.798942	0.72058177	7.111390
Canada	west	86.366181	-1.25149570	10.540653
Chile	latin	54.261364	-0.37280241	9.787202

1-10 of 75 rows

Previous 1 2 3 4 5 6 ... 8 Next

skim()

Display summary statistics

```
library(skimr)  
skim(flfp)
```



The data set

```
library(skimr)

skim(flfp)
```

— Data Summary —

	Values
Name	flfp
Number of rows	44670
Number of columns	15

Column type frequency:

character	1
factor	7
numeric	7

Group variables	None
-----------------	------

— Variable type: character —

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
1 year	0	1	4	4	0	10	0

— Variable type: factor —

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
1 cntry	0	1	FALSE	78	Sou: 1483, Ind: 1312, Ira: 1174, Jap: 1052
2 denom	1411	0.968	FALSE	4	Chr: 18632, Mus: 12567, Non: 7131, Oth: 4929
3 age_gr	125	0.997	FALSE	6	26-: 11959, 36-: 10847, 46-: 8435, 18-: 7086
4 edu	705	0.984	FALSE	3	Mid: 21769, Low: 14547, Hig: 7649
5 marit	94	0.998	FALSE	3	Mar: 30668, Sin: 8257, Div: 5651
6 children	608	0.986	FALSE	4	2-3: 18755, No : 9784, 1 c: 8253, 4 a: 7270
7 region	0	1	FALSE	7	Cen: 7123, Eas: 6981, Sou: 6754, MEN: 6428

— Variable type: numeric —

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
1 wgt	0	1	0.981	0.380	0.0574	0.877	1	1	5	
2 lfp	0	1	0.657	0.475	0	0	1	1	1	
3 patr_values	2836	0.937	-0.0171	0.984	-3.20	-0.796	-0.0752	0.648	5.56	
4 religious	7	1.00	0.431	0.495	0	0	0	1	1	
5 patr_mean	621	0.986	0.112	0.984	-2.08	-0.793	0.120	0.879	1.86	
6 log_gdp	998	0.978	9.41	1.03	6.70	8.72	9.51	10.3	11.7	
7 muslim_cntry	0	1	0.303	0.460	0	0	0	1	1	

A bit wierd that year is stored as a character variable?

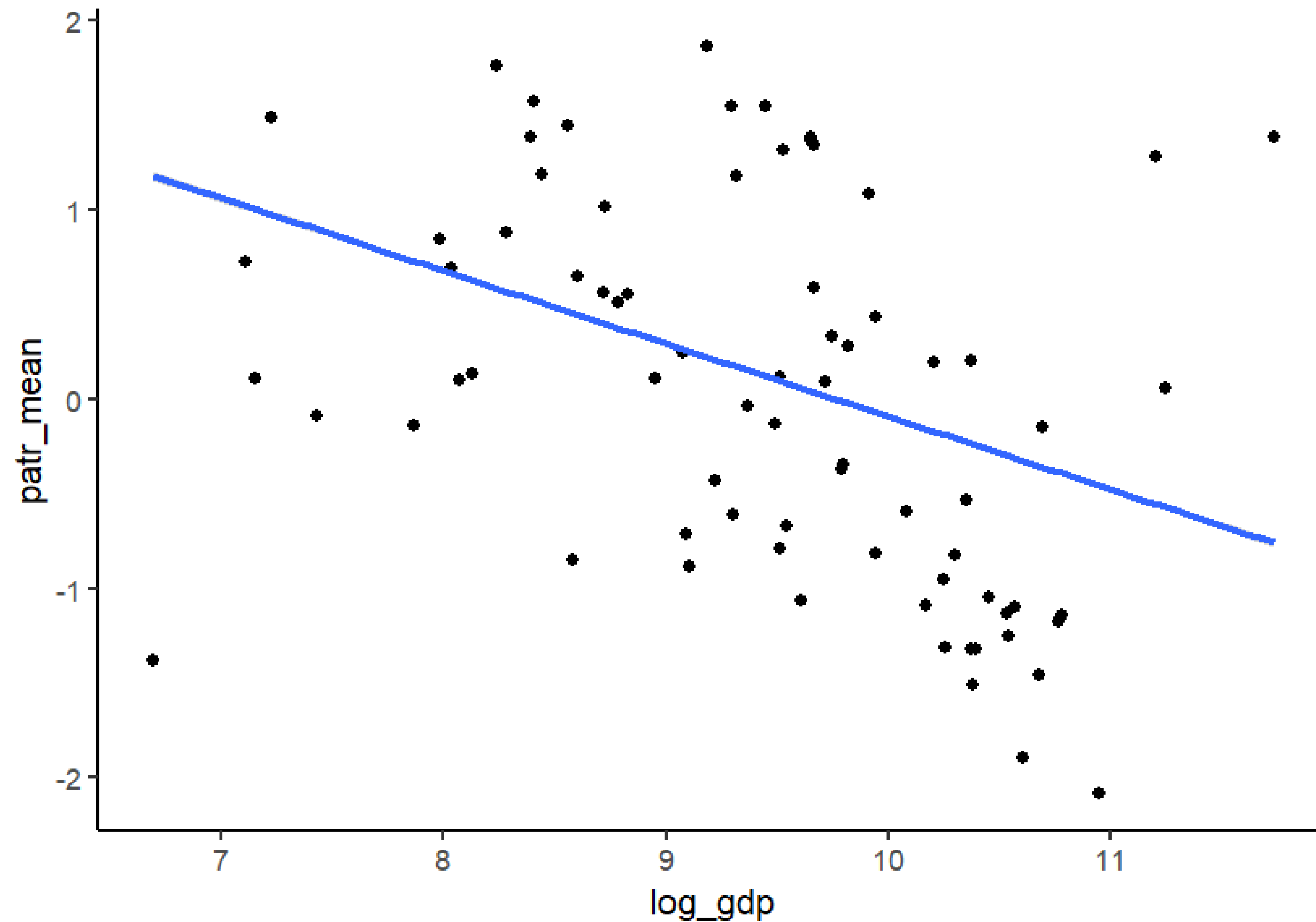
Convert it to numeric or integer!

```
flfp <- type_convert(flfp)
#or
flfp$year <- as.integer(flfp$year)
```

Notice type_convert will give different results from type.convert

Im()

```
ggplot(flfp_agg, aes(x = log_gdp, y = patr_mean)) +  
  geom_point() +  
  geom_smooth(method = "lm" )
```



formulas

Formula only needs to include the response/dependent variable and predictors/(independent variables & controls)

$$y = \alpha + \beta x + \epsilon$$

$y \sim x$

Response

A tilde

The predictors

lm()

Fit a linear model to data

```
my_model <- lm(pat_r_mean ~ log_gdp, data = flfp_agg)
```

A formula that
describes the model
equation

The data set

Your Turn

Fit the model below and then examine the output. What does it look like?

```
my_model <- lm(pat_r_mean ~ log_gdp, data = flfp_agg)
```

my_model

Call:

```
lm(formula = patr_mean ~ log_gdp, data =  
flfp_agg)
```

Coefficients:

(Intercept)	log_gdp
3.4488	-0.3621

Summary!

```
summary(my_model)
```

Call:

```
lm(formula = patr_mean ~ log_gdp, data = flfp_agg)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.4011	-0.7323	-0.1004	0.6470	2.1837

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.4488	0.9818	3.513	0.000765 ***
log_gdp	-0.3621	0.1034	-3.504	0.000788 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

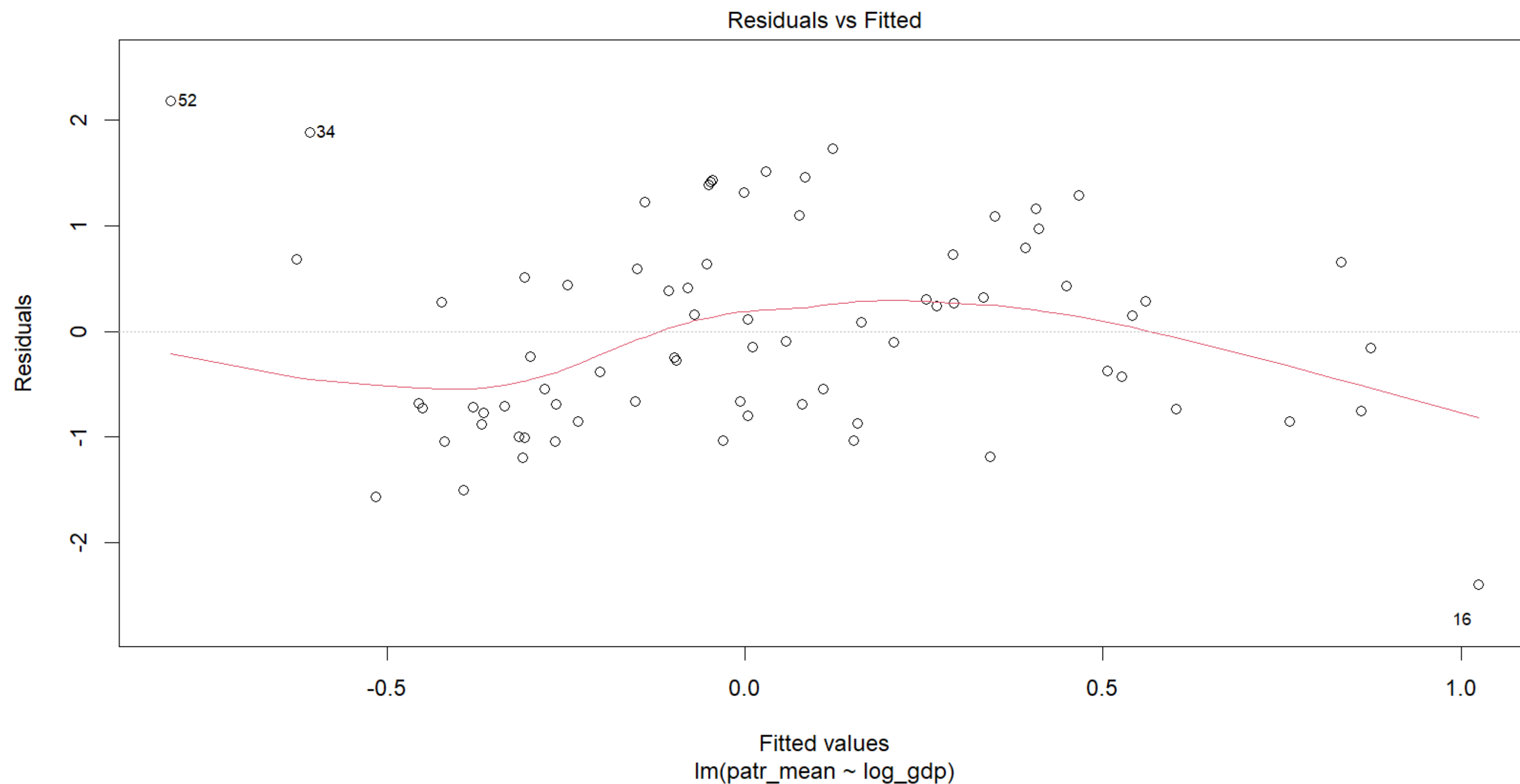
Residual standard error: 0.9492 on 73 degrees of freedom

Multiple R-squared: 0.144, Adjusted R-squared: 0.1322

F-statistic: 12.28 on 1 and 73 DF, p-value: 0.0007875

Model diagnostics

```
plot(my_model, which = 1)
```



Regression Assumptions/Issues

Assumption	When to check	How to test	Violating this assumption implies	How to Respond
<u>1) Linear relationship between Y and the X terms</u>	Check this first, during your bivariate checks.	Scatterplot with regression line. Correlation coefficient (Pearson's R).	Biased coefficients	Transformation of an X into ordinal form. Adding an X-squared term. Transform your Y variable. (This assumption is not problematic if the residuals are homoscedastic and normally distributed (points 4 and 5)).
<u>2) Low multicollinearity</u>	Can do initial check in bivariate analysis if you like, (correlations) but def. check this in the coefficient table for your multiple regression.	Low Pearson's R in X correlations (.8 or smaller) You want VIF less than 3. 5 is potential problem. 10 is a violation. (Ignore this in interaction terms)	Biased p-values (inflated confidence intervals and significance)	Remove a variable or combine two into an index. (If you are only predicting (e.g. Machine learning), do nothing.)
3) Normal distribution of errors (residuals)	Check this when you find a model you are beginning to like.	P-P plot & residuals histogram (regression-plots window) Or Q-Q plot (more tricky)	Biased p-values	In large samples, not a worry
4) Homoscedasticity (constant variance) of errors (residuals) across all values of X	Check this when you find a model you are beginning to like.	Scatterplot of residuals by predicted value (Y: zresid, X:zpred), (regression-plots window) Avoid 'funneling'	Biased p-values	In large samples, not necessarily a worry. Redefine variables with large ranges (e.g. raw income) Y transformation (e.g. box-cox transformation, log transformation) Switch to a different kind of regression (read Williams et al.)
5) Errors should be independent from one another	Check this when you find a model you are beginning to like.	Durbin-Watson (1.5 to 2.5 is normal. Close to 0 or 4 is problematic) (regression-statistics window)	Biased p-values	Depends on type of dependence (e.g. time series model or multi-level regression. There is some major grouping variable you are probably missing, e.g. country)
6) Outliers	Check this when you find a model you are beginning to like.	Cook's distance (you want smaller than .5, definitely smaller than 1) (regression-save window) Casewise diagnostics(regression-	Biased coefficients	Doublecheck data accuracy. In a big sample, usually do nothing.
See Williams et. al 2013				

broom



Turns model output into data frames

```
# install.packages("tidyverse")  
library(broom)
```

broom

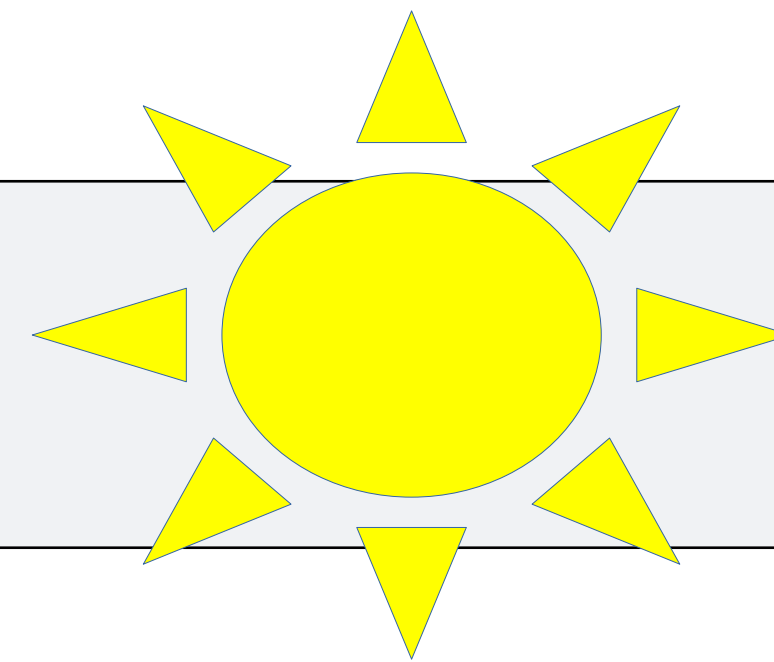
Broom includes three functions which work for most types of models (and can be extended to more):

1. **tidy()** - returns model coefficients, stats
2. **glance()** - returns model diagnostics
3. **augment()** - returns predictions, residuals, and other raw values

tidy()

Returns useful **model output** as a data frame (can be handy for prepping for publication)

```
tidy(my_model)
```



A tibble: 2 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	3.45	0.982	3.51	0.000765
2 log_gdp	-0.362	0.103	-3.50	0.000788

glance()

Returns common **model diagnostics** as a data frame

```
glance(my_model)
```

A tibble: 1 × 12

	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.144	0.132	0.949	12.3	0.000788	1	-102.	209.	216.

augment()

Returns data frame of **model output related to original data points**

```
augment(my_model)
```

```
# A tibble: 75 × 8 predictions
  patr_mean log_gdp .fitted .resid .hat .sigma .cooksd .std.resid
    <dbl>   <dbl>   <dbl>   <dbl> <dbl> <dbl>   <dbl>   <dbl>
1     1.32     9.53 -0.00152  1.32  0.0134  0.943  0.0133    1.40
2     0.552     8.83  0.253    0.299  0.0178  0.955  0.000917   0.318
3    -1.13    10.5  -0.365   -0.769  0.0274  0.951  0.00951  -0.821
4     1.33     9.66 -0.0510  1.39  0.0139  0.942  0.0153    1.47
5     0.0901    9.72 -0.0700  0.160  0.0142  0.956  0.000209   0.170
6    -0.793    9.51  0.00394  -0.797  0.0134  0.951  0.00485  -0.845
7    -0.434    9.22  0.109   -0.543  0.0139  0.954  0.00234  -0.576
8     0.721    7.11  0.874   -0.153  0.0776  0.956  0.00119  -0.168
9    -1.25    10.5  -0.368   -0.883  0.0277  0.950  0.0127  -0.944
10   -0.373    9.79 -0.0953  -0.277  0.0148  0.955  0.000650  -0.294
# i 65 more rows
# i Use `print(n = ...)` to see more rows
```

Your turn

- Using `moral_issues`, predict `prop_lib` (as Y) with `prop_cons` (X) across all issues and years. Use `lm()` to model this. Then use `tidy`, `glance`, and `plot` functions to:
 1. Extract the estimated regression coefficients.
 2. Look at the estimates of model fit.
 3. Make diagnostics plots.
- Extra: Use `select` to extract only `adj.r.squared` and `BIC` from the model fit data frame.

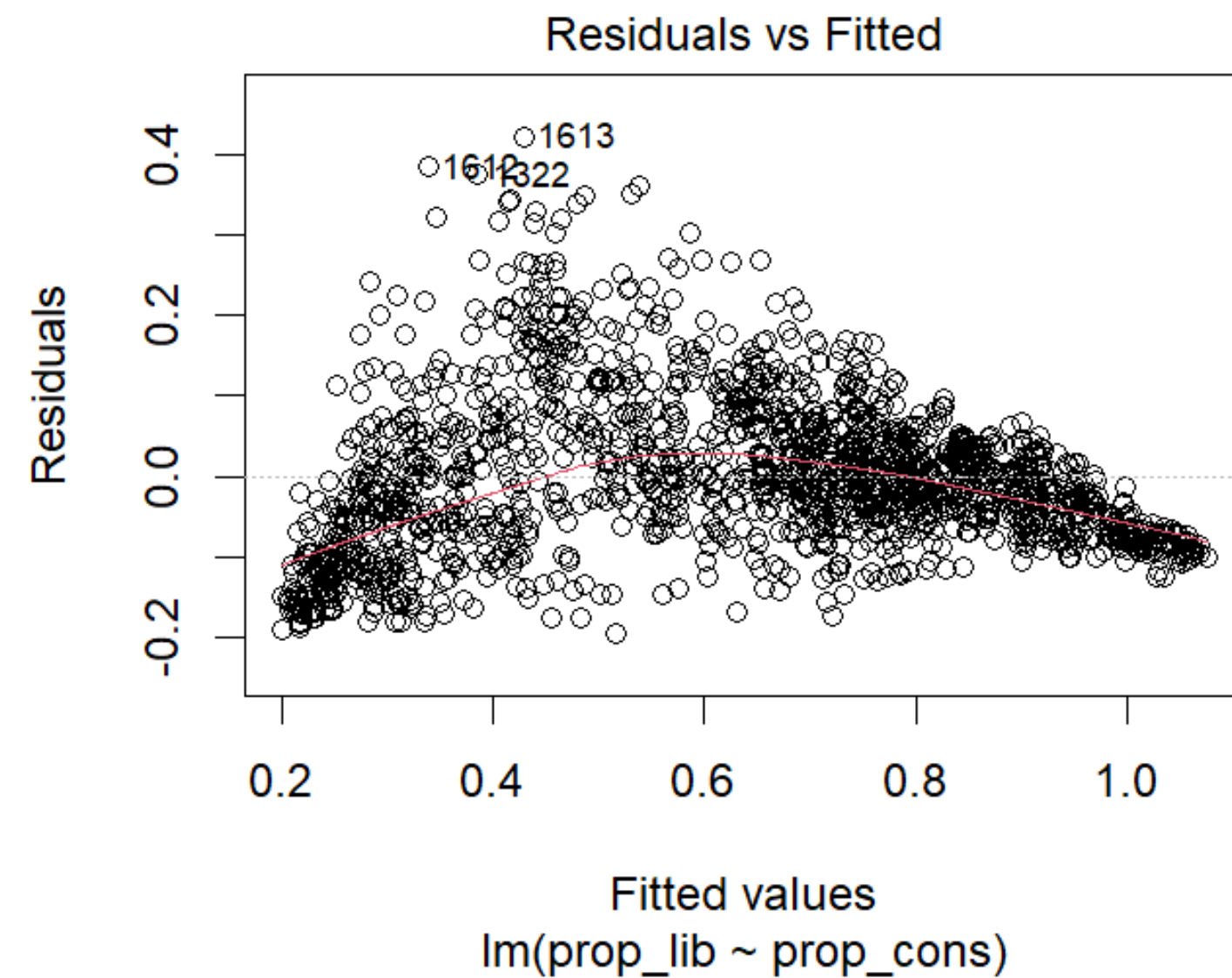
```
issue_pairing <- lm(prop_lib ~ prop_cons,  
data=moral_issues )
```

```
tidy(issue_pairing)
```

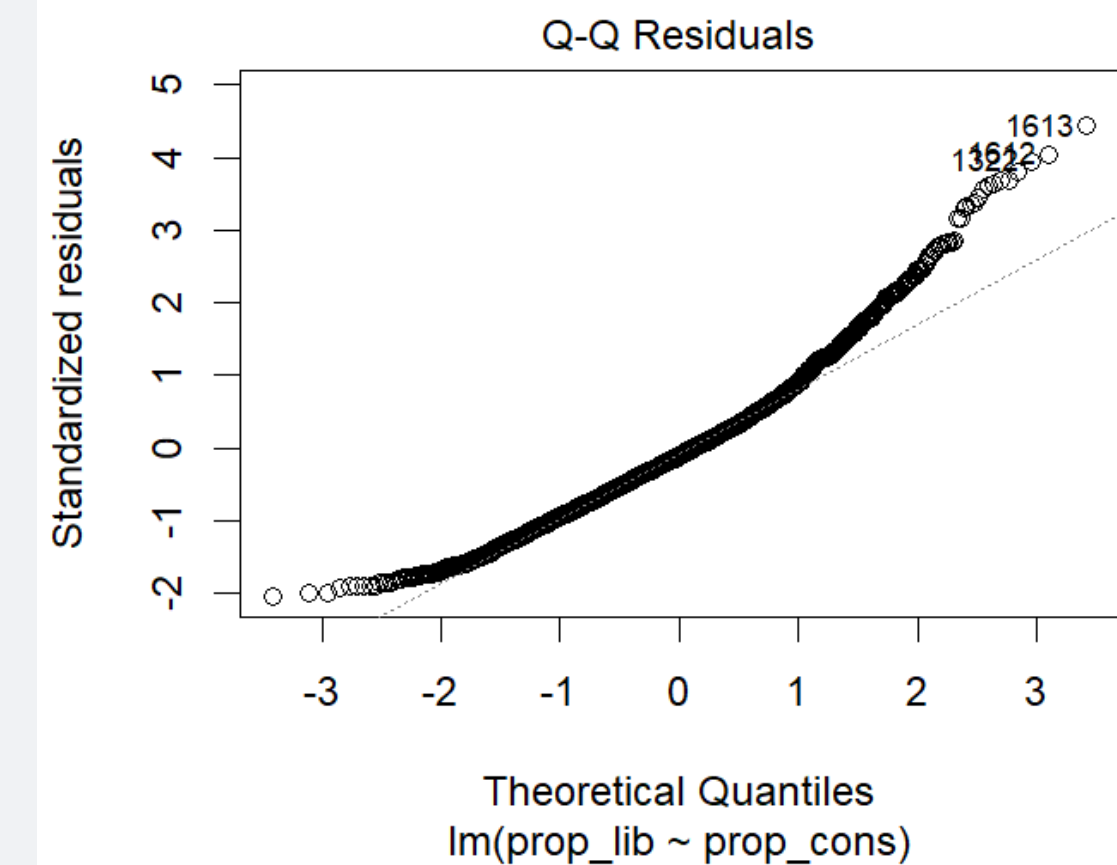
```
A tibble: 2 × 5  
  term          estimate std.error statistic    p.value  
  <chr>         <dbl>      <dbl>      <dbl>    <dbl>  
(Intercept)    0.183    0.00505      36.3 5.02e-211  
prop_cons      0.903    0.00907      99.5 0
```

```
glance(mod_gdp)
```

```
# A tibble: 1 × 12  
  r.squared adj.r.squared sigma statistic p.value    df logLik   AIC    BIC deviance  
    <dbl>      <dbl>    <dbl>      <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>  
1    0.861    0.861 0.0951      9904.      0     1  1493. -2980. -2964.    14.4  
# i 2 more variables: df.residual <int>, nobs <int>
```



```
plot(issue_pairing
, which = 1)
```



```
plot(issue_pairing,
which = 2)
```

```
glance(issue_pairing) %>%
  select(adj.r.squared, BIC)
```

```
# A tibble: 1 × 2
  adj.r.squared    BIC
      <dbl>    <dbl>
1      0.861 -2964.
```

Multivariate regression

To fit multiple predictors,
Simply add multiple variables to the formula
with a + sign entered in the lm() function:

```
wvs_flfp ~ patr_mean + log_gdp
```

Your turn

- Predict `wvs_flfp` using both `patr_mean` and `log_gdp`.
- Call up the coefficients for the model using `tidy()` and the adjusted R squared and BIC using `glance()`

```
mod_agg <- lm(wvs_flfp ~ patr_mean + log_gdp, data =  
flfp_agg)  
tidy(mod_agg)
```

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	56.305623	19.789883	2.8451720	5.776306e-03
patr_mean	-13.732938	2.181985	-6.2937819	2.152952e-08
log_gdp	1.198427	2.082481	0.5754802	5.667608e-01

3 rows

```
glance(mod_agg) %>%  
  select(adj.r.squared, BIC)
```

adj.r.squared <dbl>	BIC <dbl>
0.3927989	658.0563

1 row

Quadratic terms

Quadratic regression

```
mod_agg_quadratic <- lm(wvs_flfp ~ patr_mean + log_gdp +  
I(log_gdp^2), data = flfp_agg)  
tidy(mod_agg_quadratic)
```

Keep both!

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	177.574920	122.626019	1.4481015	1.519904e-01
patr_mean	-13.364826	2.212630	-6.0402443	6.388181e-08
log_gdp	-25.659897	26.883594	-0.9544816	3.430784e-01
I(log_gdp^2)	1.465696	1.462667	1.0020708	3.197124e-01

4 rows

```
glance(mod_agg) %>%
  select(adj.r.squared, BIC)
```

adj.r.squared	BIC
<dbl>	<dbl>
0.3928338	661.3205

1 row

```
glance(mod_agg_quadratic) %>%
  select(adj.r.squared, BIC)
```

adj.r.squared	BIC
<dbl>	<dbl>
0.3927989	658.0563

1 row

Your turn

- Model `wvs_flfp` against `patr_mean` using a quadratic term in the regression. Keep `log_gdp` as a control variable with a linear effect. Does the quadratic term of `patriarchal values` improve the model fit?

```
mod_agg_patr2 <- lm(wvs_flfp ~ patr_mean + I(patr_mean^2) + log_gdp,
                    data = flfp)

tidy(mod_agg_patr2)
```

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	51.136783	18.838272	2.714516	8.325934e-03
patr_mean	-12.907403	2.086458	-6.186276	3.508017e-08
I(patr_mean^2)	-6.101634	2.020532	-3.019815	3.513014e-03
log_gdp	2.406118	2.014244	1.194551	2.362378e-01

4 rows

```
glance(mod_agg_patr2) %>%
  select(adj.r.squared, BIC)
```

adj.r.squared	BIC
<dbl>	<dbl>
0.4543326	653.3111

1 row

```
glance(mod_agg_quadratic) %>%
  select(adj.r.squared, BIC)
```

adj.r.squared	BIC
<dbl>	<dbl>
0.3927989	658.0563

1 row

Categorical predictors

Regional differences in FLFP

```
mod_reg <- lm(wvs_flfp ~ region, data = flfp)
tidy(mod_reg)
```

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	86.79406	4.785285	18.137699	9.352547e-28
regions/est europe	-4.38315	6.521244	-0.672134	5.037758e-01
regionlatin	-25.74192	7.566200	-3.402225	1.123613e-03
regioneast asia	-15.52003	7.097725	-2.186620	3.221460e-02
regionss africa	-14.10942	7.309644	-1.930248	5.774795e-02
regioncentr asia	-40.46311	7.097725	-5.700857	2.810411e-07
regionmena	-42.94064	6.767415	-6.345205	2.099530e-08

7 rows

Regional differences in FLFP

```
mod_reg <- lm(wvs_flfp ~ region, data = flfp)
tidy(mod_reg)
```

Where is the West?

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	86.79406	4.785285	18.137699	9.352547e-28
regions/est europe	-4.38315	6.521244	-0.672134	5.037758e-01
regionlatin	-25.74192	7.566200	-3.402225	1.123613e-03
regioneast asia	-15.52003	7.097725	-2.186620	3.221460e-02
regionss africa	-14.10942	7.309644	-1.930248	5.774795e-02
regioncentr asia	-40.46311	7.097725	-5.700857	2.810411e-07
regionmena	-42.94064	6.767415	-6.345205	2.099530e-08

7 rows

Interaction terms

Interaction model

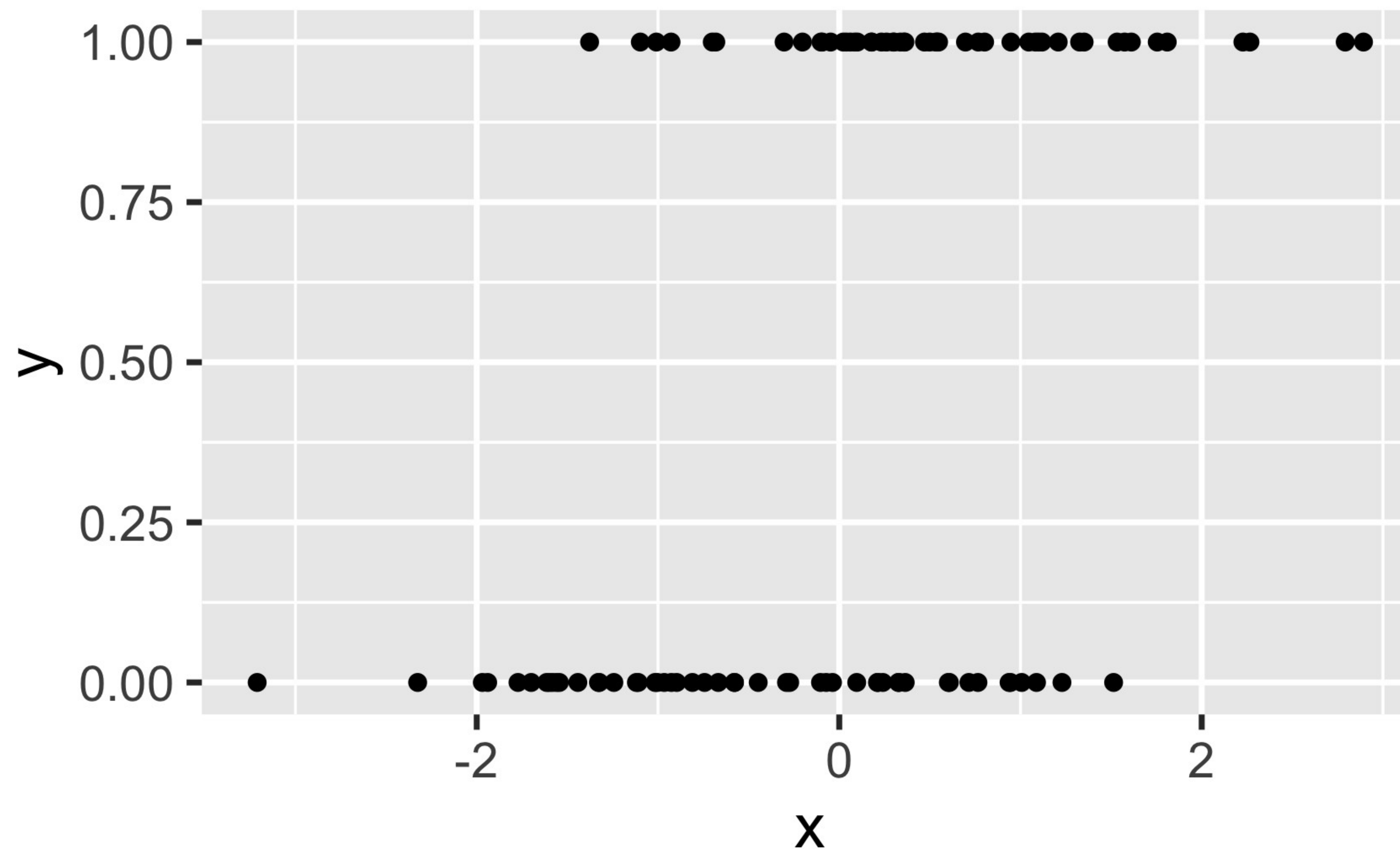
```
mod_int <- lm(wvs_flfp ~ patr_mean + muslim + patr_mean:muslim,  
data = flfp_agg)
```

```
tidy(mod_int)
```

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	73.6	2.65	27.7	8.47e-40
2 patr_mean	-6.14	2.96	-2.07	4.18e- 2
3 muslimTRUE	6.72	10.1	0.665	5.09e- 1
4 patr_mean:muslimTRUE	-23.2	8.43	-2.75	7.61e- 3

glm()

Binary outcome



flfp

cntry <fctr>	lfp <dbl>	patr_values <dbl>	patr_mean <dbl>	denom <fctr>	age_gr <fctr>	religious <dbl>
Andorra	1	-0.42782837	-1.5438549	Christ	36-45	0
Andorra	1	0.33482220	-1.5438549	Christ	18-25	0
Andorra	1	0.33482220	-1.5438549	Christ	26-35	0
Andorra	1	0.33482220	-1.5438549	None	18-25	0
Andorra	1	1.86012335	-1.5438549	Christ	36-45	0
Andorra	1	-0.42782837	-1.5438549	Christ	56-65	0
Andorra	1	-0.42782837	-1.5438549	None	26-35	0
Andorra	1	4.14807507	-1.5438549	None	>66	0
Andorra	1	-1.19047894	-1.5438549	None	56-65	0
Andorra	1	-1.19047894	-1.5438549	None	26-35	0

1-10 of 44,670 rows

Previous **1** 2 3 4 5 6 ... 100 Next

Skim summary statistics

n obs: 44670

n variables: 15

— Variable type:character

variable	missing	complete	n	min	max	empty	n_unique
year	0	44670	44670	4	4	0	10

— Variable type:factor

variable	missing	complete	n	n_unique	top_counts	ordered
age_gr	125	44545	44670	6	26-: 11959, 36-: 10847, 46-: 8435, 18-: 7086	FALSE
children	608	44062	44670	4	2-3: 18755, No : 9784, 1 c: 8253, 4 a: 7270	FALSE
cntry	0	44670	44670	78	Sou: 1483, Ind: 1312, Ira: 1174, Jap: 1052	FALSE
denom	1411	43259	44670	4	Chr: 18632, Mus: 12567, Non: 7131, Oth: 4929	FALSE
edu	705	43965	44670	3	Mid: 21769, Low: 14547, Hig: 7649, NA: 705	FALSE
marit	94	44576	44670	3	Mar: 30668, Sin: 8257, Div: 5651, NA: 94	FALSE
region	0	44670	44670	7	Cen: 7123, Eas: 6981, Sou: 6754, MEN: 6428	FALSE

— Variable type:numeric

variable	missing	complete	n	mean	sd	p0	p25	p50	p75	p100	hist
lfp	0	44670	44670	0.66	0.47	0	0	1	1	1	
log_gdp	998	43672	44670	9.41	1.03	6.7	8.72	9.51	10.25	11.74	
muslim_cntry	0	44670	44670	0.3	0.46	0	0	0	1	1	
patr_mean	621	44049	44670	0.11	0.98	-2.08	-0.79	0.12	0.88	1.86	
patr_values	2836	41834	44670	-0.017	0.98	-3.2	-0.8	-0.075	0.65	5.56	
religious	7	44663	44670	0.43	0.5	0	0	0	1	1	
wgt	0	44670	44670	0.98	0.38	0.057	0.88	1	1	5	

glm()

Fits a generalised linear model to data

```
model_val <- glm(lfp ~ patr_values + cntry,  
                 family = binomial(link = "logit"),  
                 data = flfp_ind)
```

**Modelled
distribution**

Link function

Coefficients

```
glm(lfp ~ patr_values + cntry,  
    family = binomial(link = "logit"),  
    data = flfp_ind)  
  
tidy(mod_val) %>% filter(!str_detect(term, "cntry"))  
  
## # A tibble: 2 x 5  
##   term          estimate std.error statistic    p.value  
##   <chr>          <dbl>      <dbl>      <dbl>    <dbl>  
## 1 (Intercept)   -0.333     0.110     -3.02  2.57e- 3  
## 2 patr_values  -0.289     0.0119    -24.2  1.13e-129
```

*Avg.
Change
in log
odds.*

Odds ratios

```
tidy(mod_val, exponentiate = TRUE) %>%  
  filter(!str_detect(term, "cntry"))
```

```
## # A tibble: 2 x 5
```

##	term	estimate	std.error	statistic	p.value
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	(Intercept)	0.717	0.110	-3.02	2.57e- 3
## 2	patr_values	0.749	0.0119	-24.2	1.13e-129

*Odds ratio
compared to
reference
category.*

Your turn

- Read the individual level data.
- Add the variable `patr_mean` to the model we looked at last and remove `cntry`.
- Which values have stronger effect: the individual or the country mean?

```
mod_val_cntr <- glm(lfp ~ patr_values + patr_mean ,  
                    family = binomial(link = "logit"),  
                    data = flfp)
```

```
tidy(mod_val_cntr, exponentiate = TRUE)
```

```
# A tibble: 3 × 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	3.23	0.0849	13.8	2.83e- 43
2	patr_values	0.748	0.0120	-24.1	9.32e-129
3	patr_mean	0.319	0.101	-11.3	1.91e- 29

Categorical variables

```
mod_val_edu <- glm(lfp ~ patr_values + edu + cntry,  
                  family = binomial(link = "logit"),  
                  data = flfp  
  
tidy(mod_val_edu, exponentiate = TRUE) %>%  
  filter(!str_detect(term, "cntry"))  
  
## # A tibble: 4 x 5  
##   term          estimate std.error statistic    p.value  
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>  
## 1 (Intercept)    0.452     0.116     -6.84 7.80e- 12  
## 2 patr_values    0.827     0.0125    -15.2 5.78e- 52  
## 3 eduMiddle      2.49      0.0300     30.4 6.54e-203  
## 4 eduHigh        7.58      0.0458     44.2 0.
```

Your turn

- Estimate *lfp* against country *region* (s) on the individual level. Add *edu*, *age_gr*, *marit*, *children*, *religious*, and *denom* as control variables, but *do not include* *cntry*. Compare the odds of a female from a MENA country to be employed to those of a female from the West?

```
mod_reg <- glm(lfp ~ region + edu + age_gr + marit +  
               children + religious + denom,  
               family = binomial(link = "logit"),  
               data = flfp)  
  
tidy(mod_reg, exponentiate = TRUE)
```





term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>
(Intercept)	4.9322259	0.08839417	18.053118
regionSouth/Eastern Europe	0.8798559	0.05377818	-2.380095
regionLatin America	0.3681781	0.05297530	-18.861403
regionEastern Asia	0.6798980	0.05277278	-7.310824
regionSub-Saharan Africa	1.1745315	0.05661489	2.841467
regionCentr/South/Western Asia	0.2485516	0.05441782	-25.581778
regionMENA	0.3651798	0.06225509	-16.181253
eduMiddle	1.9280904	0.02774645	23.661772
eduHigh	6.0040955	0.04436911	40.398417
age_gr18-25	2.1752846	0.07509413	10.349137

1-10 of 23 rows | 1-4 of 5 columns

Previous

1

 2
 3
 Next

Your turn

- Add `patr_values` and `patr_mean` to the model we just fit. How does the regional differences change?

```
mod_reg_val <- glm(lfp ~ region + patr_values + patr_mean +  
                    edu + age_gr + marit + children +  
                    religious + denom,  
                    family = binomial(link = "logit"),  
                    data = flfp)  
  
tidy(mod_reg_val, exponentiate = TRUE)
```





term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>
(Intercept)	3.2440468	0.10573128	11.130306
regionSouth/Eastern Europe	1.1982986	0.06231925	2.902839
regionLatin America	0.4407688	0.05870425	-13.955291
regionEastern Asia	1.2392828	0.07599073	2.823145
regionSub-Saharan Africa	2.1419561	0.07917011	9.621301
regionCentr/South/Western Asia	0.5084256	0.08494248	-7.963464
regionMENA	0.7235934	0.09291332	-3.482016
patr_values	0.8836590	0.01288577	-9.598496
patr_mean	0.6964265	0.03230143	-11.200529
eduMiddle	1.9417538	0.02945170	22.531519

Generalised linear models

- Use different link functions to connect variety of outcomes to the linear predictor.

```
glm(y ~ x, family = poisson(link = "log"))
```

- Check the full list with ?family.